

AmritFlow Architecture & Data Sovereignty Specification

Technical Brief v1.0 — Sovereign Mailbox Intelligence & Autonomous Triage Engine

100% ON-PREMISE SOVEREIGN

MULTI-DOMAIN AGENTIC INFRA

HUMAN-IN-THE-LOOP GOVERNANCE

01. THE ENTERPRISE PROBLEM: OPERATIONAL RISK IN AP MAILBOXES

Legacy & Public Cloud Vulnerabilities

- **High Noise-to-Signal Ratios:** Unstructured AP mailboxes overwhelm finance teams, causing severe SLA drag during month-end closes.
- **Security & Fraud Risks:** Vulnerable to vendor impersonation, duplicate invoicing, and unauthorized account updates.
- **Data Sovereignty Hazards:** Traditional OCR and public cloud LLM integrations expose financial PII to external harvesting and unpredictable API latency.

AmritFlow Sovereign Solution

- **Deterministic Triage:** Automates ingestion, parsing, and data validation directly at the edge boundary.
- **Zero-Egress Security:** Ensures no financial data, invoice metadata, or vendor records leave your enterprise perimeter.
- **Autonomous Accuracy:** Sub-second extraction latency with deterministic validation against ERP master tables.

02. CORE ARCHITECTURAL PILLARS

PILLAR 01: Zero-Egress Ingestion

Local / Private VPC deployment models ensure enterprise financial data never trains third-party public models or crosses geographic boundaries. Complete data sovereignty maintained.

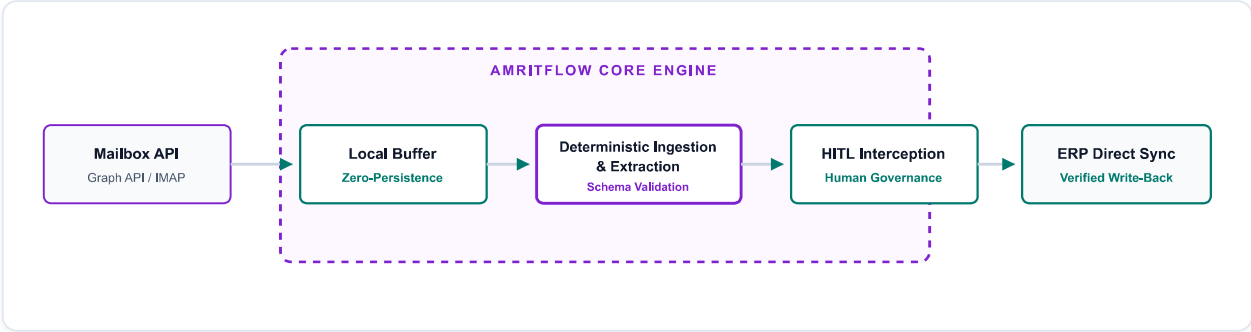
PILLAR 02: Deterministic Parsing

Combines structured extraction rules with edge ML models to convert raw email threads and PDF attachments into fully verified, schema-bound JSON packages instantly.

PILLAR 03: HITL Interception

Edge cases, threshold anomalies, and unverified vendor requests auto-route to designated human operators via an interactive dashboard before any ERP write-back occurs.

03. SYSTEM TOPOLOGY DIAGRAM



04. INFRASTRUCTURE & SECURITY SPECIFICATIONS

SPECIFICATION DOMAIN	ENTERPRISE TECHNICAL STANDARDS
Deployment Environments	RHEL 8+, Ubuntu 22.04 LTS, Windows Server 2022+, Bare-Metal Deployments, AWS/Azure Private VPCs.
Data Handling	Zero-Persistence intake layer; AES-256 encryption at rest; TLS 1.3 in transit; strict egress filtering.
ERP & Database Compatibility	REST API, Webhooks, Direct Database Connectors, Structured Batch Exports (CSV/JSON).
Governance & Compliance	Human-in-the-Loop (HITL) verification gates, role-based access control (RBAC), immutable audit logging.

Request Evaluation Access & Environment Specification

Deploy AmritFlow in your sandboxed enterprise environment for a 60-day zero-telemetry evaluation.

Visit: amritflow.com | **Direct Contact:** eval@amritflow.com